

Session 12 Quiz - Advanced Prompting

Track B | Grades 9–12 | 20 Questions

Q1 [Type: MCQ | Difficulty: Easy | QID : 53443]

A prompting technique where you give the AI examples of the kind of output you want before asking your actual question is called _____ prompting.

- A. Zero-shot
- B. Few-shot
- C. Chain
- D. Role

Correct Answer: B

Explanation: Few-shot prompting provides 2–5 examples in the prompt to show the AI the pattern you want. Zero-shot prompting gives no examples (just asks the question). Chain (chain-of-thought) prompts ask the AI to reason step-by-step. Role prompting assigns the AI a persona.

Q2 [Type: MCQ | Difficulty: Easy | QID : 53446]

A prompt that starts with "Act as a professional editor and review this paragraph..." uses a technique called _____ prompting.

- A. Token
- B. Output
- C. Role
- D. Format

Correct Answer: C

Explanation: Role prompting assigns the AI a specific persona or expertise context, which often produces more domain-appropriate responses. Token is an internal unit of text. Output is what the AI produces. Format is one element of a prompt but doesn't describe this specific technique.

Q3 [Type: MCQ | Difficulty: Easy | QID : 53449]

Asking the AI to "think step by step" before giving its final answer can improve the quality of reasoning in complex problems.

A. True

B. False

Correct Answer: A (True)

Explanation: Step-by-step reasoning (chain-of-thought prompting) often produces better results for complex problems. When the AI works through reasoning explicitly rather than jumping to a conclusion, it tends to make fewer errors on multi-step problems - particularly in math and logic. This is one of the most reliable techniques in advanced prompting.

Q4 [Type: MCQ | Difficulty: Medium | QID : 53456]

Using a persona (role-based prompting) can help a user explore different perspectives on a single topic.

A. True

B. False

Correct Answer: A (True)

Explanation: By asking the AI to act as different characters (e.g., a scientist vs. a poet, or a doctor vs. a mermaid), you can see how the same topic is explained from different viewpoints. The persona shifts the tone, vocabulary, and framing of the answer.

Q5 [Type: MCQ | Difficulty: Easy | QID : 53459]

A user provides the AI with the following instruction: "Translate the word 'Apple' into Spanish."
Which prompting technique is being utilized?

- A. Role-based prompting
- B. Chain-of-thought prompting
- C. Zero-shot prompting
- D. Few-shot prompting

Correct Answer: C

Explanation: This technique involves giving a direct instruction without any prior examples, trusting the AI's internal knowledge to complete the task. Translation of a common word is a textbook zero-shot use case.

Q6 [Type: MCQ | Difficulty: Medium | QID : 53461]

When should you ideally use zero-shot prompting?

- A. For complex logic puzzles with multiple hidden variables
- B. When the task requires a very specific output structure that the AI has never seen
- C. When you want the AI to adopt a very specific and unique creative voice
- D. For simple, well-known tasks where the AI likely has strong pre-existing knowledge

Correct Answer: D

Explanation: Zero-shot is efficient for common tasks that don't require specific formatting or complex reasoning paths. Complex logic puzzles call for chain-of-thought; unusual output structures call for few-shot; specific creative voices call for role-based or few-shot.

Q7 [Type: MCQ | Difficulty: Easy | QID : 53466]

If a user provides two examples of how to format a list before asking the AI to generate a third entry, which specific sub-type of prompting is being used?

- A. One-shot prompting
- B. Few-shot prompting
- C. Zero-shot prompting
- D. Chain-of-thought prompting

Correct Answer: B

Explanation: Few-shot prompting involves providing two or more examples to establish a clear pattern for the AI to follow. One-shot is exactly one example; zero-shot is no examples; chain-of-thought is about step-by-step reasoning.

Q8 [Type: MCQ | Difficulty: Medium | QID : 53468]

Why does role-based (persona) prompting often change the vocabulary and tone of the AI's response?

- A.** The AI accesses a different database of facts for every new character it plays
- B.** The AI is trained on millions of conversations in various voices and shifts its language to match the assigned role
- C.** It limits the AI's response to only two sentences to keep the persona consistent
- D.** It forces the AI to provide step-by-step reasoning for every answer it gives

Correct Answer: B

Explanation: By assigning a persona, you tap into specific patterns of language and knowledge the AI has learned from diverse training data. The AI is not actually remembering or believing anything - it is matching style.

Q9 [Type: MCQ | Difficulty: Medium | QID : 53471]

A student wants the AI to generate titles for short stories that always end with an exclamation mark. Which technique is most effective for teaching this unusual formatting requirement?

- A.** Chain-of-thought prompting
- B.** Zero-shot prompting
- C.** Role-based prompting
- D.** Few-shot prompting

Correct Answer: D

Explanation: Few-shot is ideal for showing the AI specific patterns or structural styles that are not standard. Showing 2–3 example titles ending with exclamation marks teaches the pattern directly - much more reliably than describing it in words.

Q10 [Type: MCQ | Difficulty: Hard | QID : 53473]

What is the primary reason for "iterating" on a prompt?

- A. To ensure the AI does not provide any reasoning for its answers
- B. The AI forgets your instructions after one minute, so you must repeat them
- C. There is no single "perfect" prompt, so you must refine your instructions to get better results
- D. You need to change the AI's persona every time you ask a new question

Correct Answer: C

Explanation: Iterating allows you to adjust constraints, context, or techniques based on the AI's initial performance. Different tasks need different combinations of techniques, and the right combination is often discovered through refinement rather than guessed correctly on the first try.

Q11 [Type: MCQ Scenario | Difficulty: Medium | QID : 53475]

A user wants to understand why people yawn. They ask: You are a friendly doctor.

Question: Which combined approach is being used in this prompt?

- A. Zero-shot and Role-based prompting
- B. Role-based prompting and Constraints
- C. Few-shot and Chain-of-thought
- D. One-shot and Persona

Correct Answer: B

Explanation: The prompt assigns a persona (friendly doctor) and sets constraints (simple words, for a five-year-old).

Q12 [Type: MCQ Scenario | Difficulty: Hard | QID : 53482]

Scenario: A user asks: "A dog, a cat, and a bird are in a house. The dog is not on the rug. The cat is on the couch. Where is the bird if it is not with the dog?"

Question: Why is chain-of-thought the best technique for this task?

- A.** It forces the AI to map out the positions of each animal logically to reach a valid conclusion
- B.** It makes the AI pretend to be one of the animals in the house
- C.** It allows the user to give examples of other logic puzzles first
- D.** It limits the AI's response to a single-word answer

Correct Answer: A

Explanation: Logic puzzles involve multiple constraints that are best handled by breaking them down into separate reasoning steps. Chain-of-thought also exposes the AI's reasoning, so if the answer is wrong you can see exactly where the logic went off-track.

Q13 [Type: FRQ - Short Answer | Difficulty: Medium | QID : 53484]

You need to quickly convert a recipe from cups to grams for baking.

Write a zero-shot prompt for this task.

Model Answer: "Convert 2 cups of flour, 1 cup of sugar, and half a cup of butter into grams." Zero-shot works because unit conversion is a simple, well-defined task - no examples needed.

Q14 [Type: FRQ - Short Answer | Difficulty: Medium | QID : 53486]

You're naming a new gaming clan with your friends. You want names that follow a specific vibe - short, sharp, and a bit edgy.

Write a few-shot prompt asking AI to generate 5 clan name ideas. Give at least 2 examples first to set the style.

Model Answer: "Here are gaming clan names I like: Nightfall, Reckless, Voidbound. Generate 5 more clan names in the same short, edgy, one-word style." Few-shot works because the examples teach AI the pattern - one word, dark vibe, no numbers or symbols.

Q15 [Type: FRQ - Short Answer | Difficulty: Medium | QID : 53487]

Look at these two prompts:

Prompt A: "Suggest 5 captions for my Instagram post."

Prompt B: "Here are captions I've used before: 'lowkey obsessed', 'main character energy', 'no thoughts just vibes'. Suggest 5 more in the same style."

(a) Which is zero-shot and which is few-shot?

(b) Which one will give you captions that actually sound like you? Why?

Model Answer: (a) Prompt A = zero-shot. Prompt B = few-shot. (b) Prompt B - because the examples show AI your specific tone (casual, short, lowercase, internet-slang). Without examples, Prompt A might return generic stuff like "Living my best life ✨" - totally off-brand.

Q16 [Type: FRQ - Scenario Analysis | Difficulty: Medium | QID : 53489]

You're trying to settle an argument with your friend about who's right in a debate (e.g., is pineapple on pizza acceptable, or should you watch movies in dub vs sub). You want AI to actually argue one side convincingly - not give a boring "both have pros and cons" answer.

Write a role-based prompt that makes AI commit to a position.

Model Answer: "Act as a passionate food critic who strongly believes pineapple does NOT belong on pizza. Give me your 3 strongest arguments against pineapple on pizza. Don't be neutral - actually defend your position with conviction and a bit of attitude." The persona forces AI to commit to a side instead of being wishy-washy.

Q17 [Type: FRQ - Scenario Analysis | Difficulty: Medium | QID : 53492]

You want to understand how a famous historical event happened - but from the perspective of someone who was actually there, not a Wikipedia summary.

Write a role-based prompt that gives AI a specific character so the explanation feels real and not textbook-style.

Model Answer: "Act as a journalist who was reporting live during the 1969 moon landing. Describe what you saw, heard, and felt that day - the atmosphere in the newsroom, the public reaction, and what it meant in the moment. Write it like a personal account, not a history lesson." The persona turns dry facts into a first-person story - way more engaging than a date-and-fact summary.

Q18 [Type: FRQ - Scenario Analysis | Difficulty: Hard | QID : 53493]

You're trying to decide whether to drop a subject/elective you're struggling with or push through it. It's not a simple yes/no - there are trade-offs. You want AI to actually think it through with you.

Write a chain-of-thought prompt that makes AI reason through the decision before giving advice.

Model Answer: "I'm thinking about dropping my coding elective because I'm struggling and my grade is hurting my overall average. But I originally picked it because I'm interested in tech as a career. Think step by step: what are the short-term benefits of dropping it? What are the long-term costs? What questions should I ask myself before deciding? End with what you'd recommend." Chain-of-thought makes AI weigh factors instead of jumping to a one-line answer.

Q19 [Type: FRQ - Scenario Analysis | Difficulty: Hard | QID : 53494]

You're writing a movie review of a film you just watched and want it to sound like a real critic - sharp opinions, specific style. You also want it to match a tone you've seen in reviews you actually like.

Write a prompt that combines role-based + few-shot techniques.

Model Answer: "Act as a sharp, no-nonsense movie critic who doesn't hold back. Here are 2 review lines in the style I want: (1) 'The plot was thinner than the popcorn bag I held it in.' (2) 'Three hours of my life I'd rather use to watch paint dry - at least paint commits to a color.' Now write a 100-word review of the latest superhero movie I just watched (you can pick any recent one). Match the same witty, brutal style." Combines role (critic) + few-shot (style samples) for a result with both expertise and exact tone.

Q20 [Type: FRQ - Scenario Analysis | Difficulty: Hard | QID : 53497]

You're trying to figure out the best phone to buy under ₹25,000. You care about camera, battery, and gaming performance. You don't want a generic "top 5" listicle - you want AI to actually reason through it like a tech-savvy friend would.

Write a prompt that combines role-based + chain-of-thought techniques.

Model Answer: "Act as a tech reviewer who has personally tested every major phone in the budget category. I want to buy a phone under ₹25,000 that's great for photography, lasts a full day on battery, and handles games like BGMI smoothly. Think step by step: (1) What are the top 3 phones currently in this budget range? (2) Compare their cameras - which one wins and why? (3) Compare their battery life under heavy use. (4) Compare their gaming performance. (5) Based on all of the above, give me your final recommendation." Combines role (tech reviewer = expertise) + chain-of-thought (structured reasoning = real comparison) so you get a recommendation you can actually trust.